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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 1.581600024152583, median 1.4880997250983514, std: 2.167782300321561
Gyroscope error (imu0):        mean 1.4703531990184484, median 1.1555997605279769, std: 1.708094430210849
Accelerometer error (imu0):    mean 0.6519653248011821, median 0.48526871749159317, std: 0.8169022496631232

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 1.581600024152583, median 1.4880997250983514, std: 2.167782300321561
Gyroscope error (imu0) [rad/s]:     mean 0.12487848711751581, median 0.09814618005009468, std:
0.14507007461947224
Accelerometer error (imu0) [m/s^2]: mean 0.357420515574372, median 0.2660340797278626, std:
0.44784225807944145

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0):
[[ 0.01203722 -0.99959525  0.02577684  0.00311205]
 [ 0.99991565  0.01215886  0.00456729 -0.02695889]
 [-0.00487886  0.02571968  0.99965729 -0.01496471]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.01203722  0.99991565 -0.00487886  0.02684614]
 [-0.99959525  0.01215886  0.02571968  0.00382347]
 [ 0.02577684  0.00456729  0.99965729  0.01500249]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.10510727524125711


Gravity vector in target coords: [m/s^2]
[-0.15356529 -9.80534744  0.00150169]
```

Calibration configuration
==========================

cam0

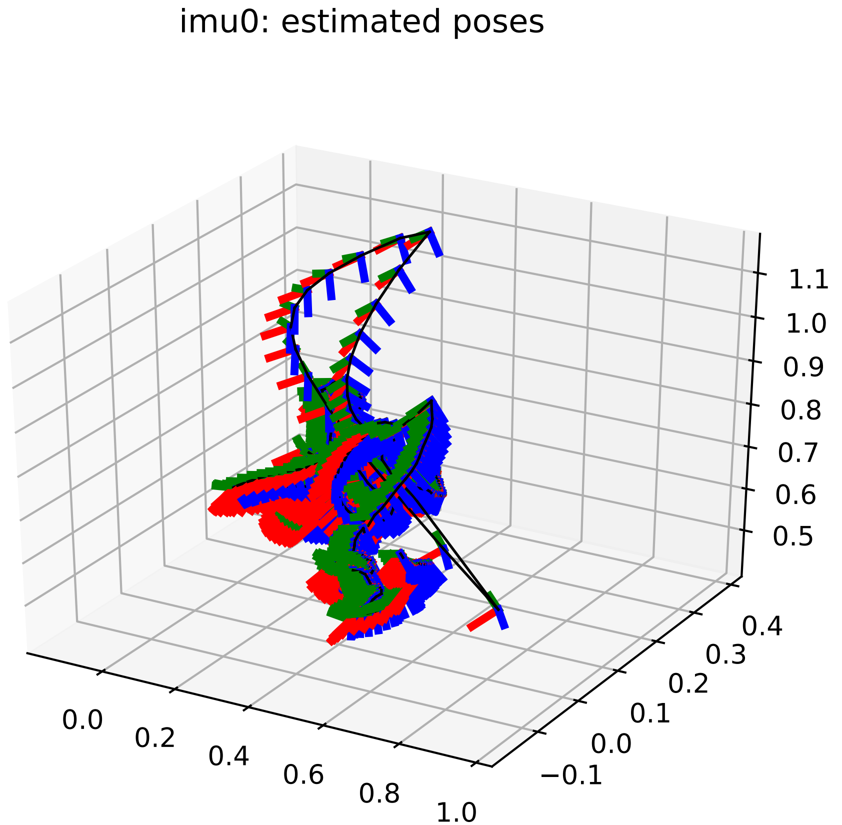
Camera model: pinhole
Focal length: [567.2852783486742, 566.4961372581788]
Principal point: [652.7255421321942, 397.4199032103684]
Distortion model: equidistant
Distortion coefficients: [-0.05551145786489933, 0.029442090222052172, -0.028299846668714417, 0.00781929524076502]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

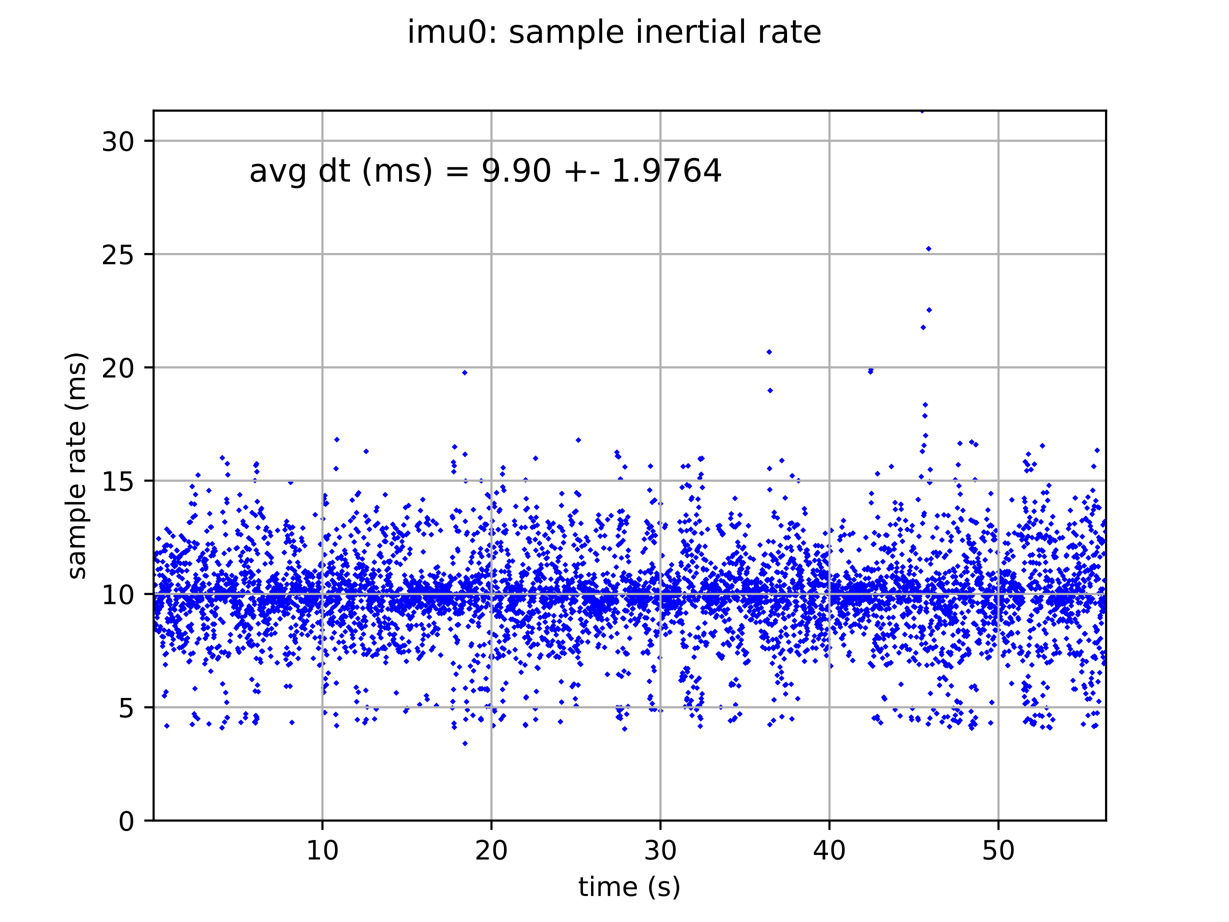
IMU configuration
=================

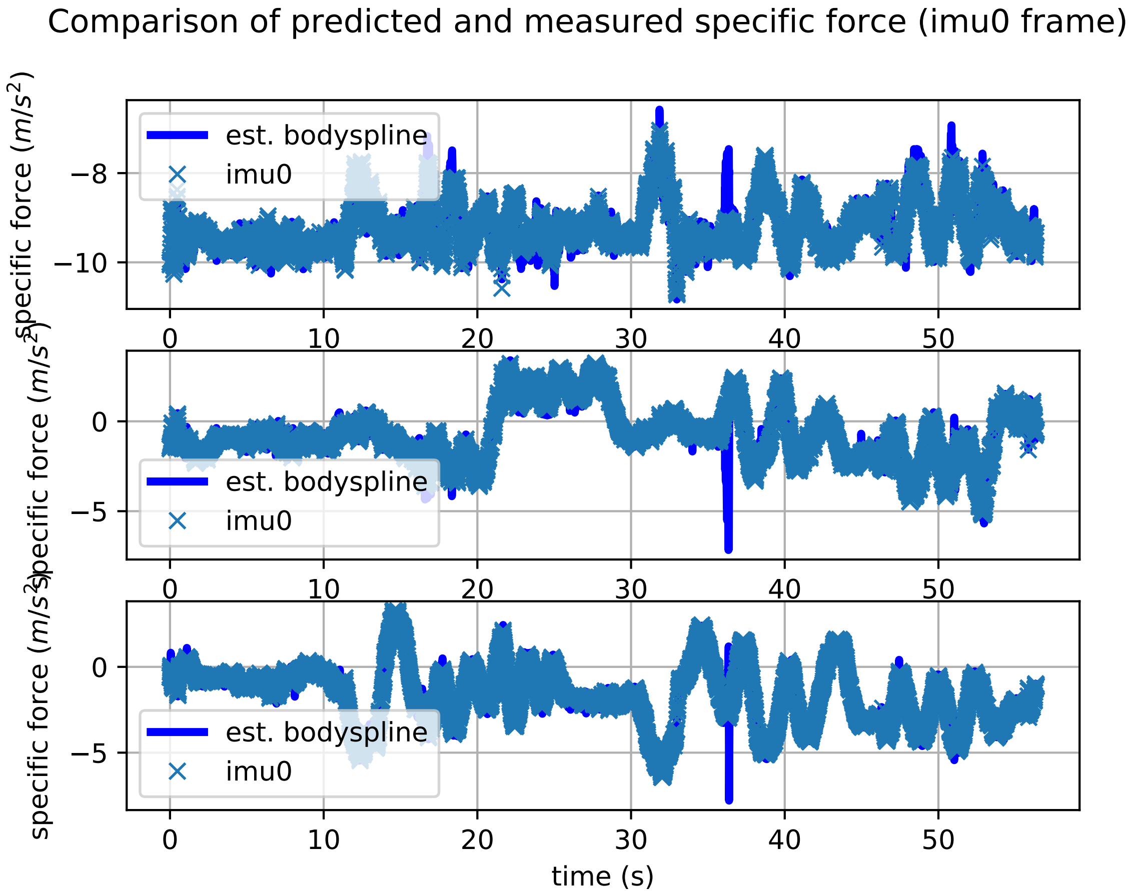
IMU0:

Model: scale-misalignment
Update rate: 100
Accelerometer:
Noise density: 0.05482201307
Noise density (discrete): 0.5482201307
Random walk: 0.04369065701339204
Gyroscope:
Noise density: 0.00849309453
Noise density (discrete): 0.08493094529999999
Random walk: 0.00011002624872473468

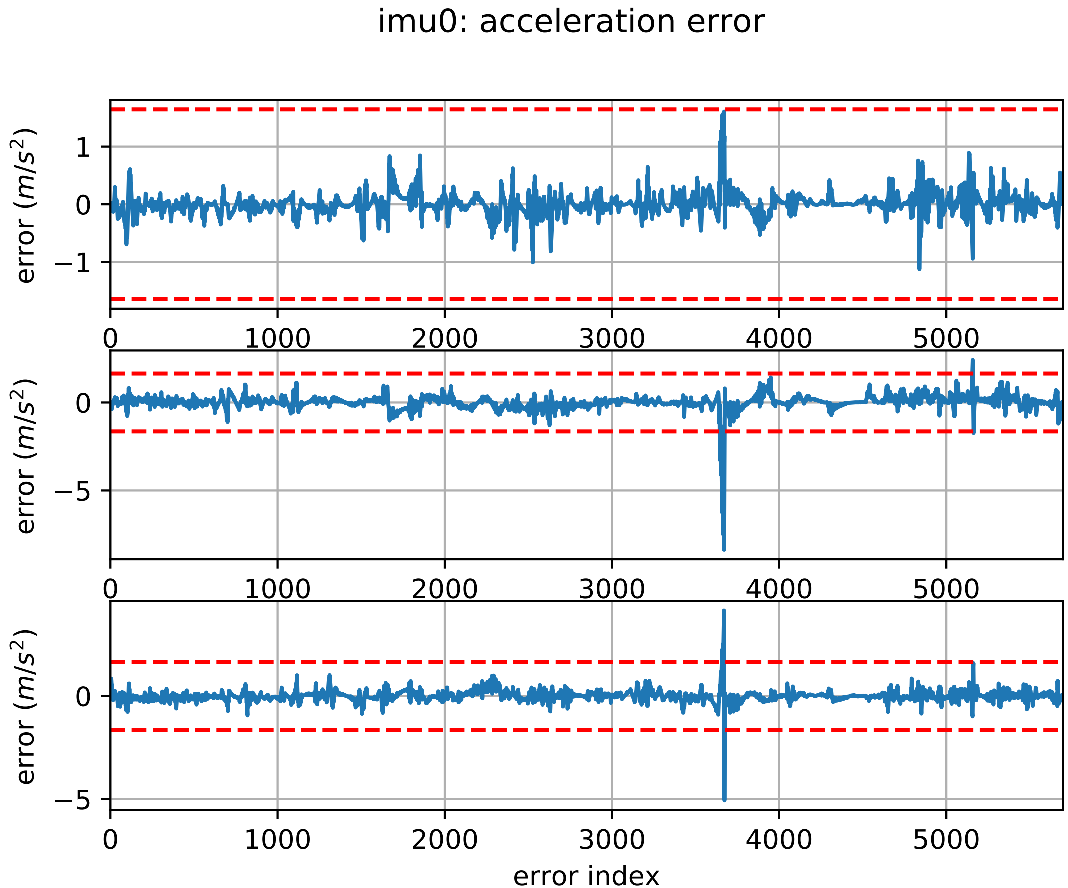
```
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
Gyroscope:
M:
[[ 0.96141986  0.          0.        ]
[ 0.05110385  1.0117608   0.        ]
[-0.14792741  0.03688996  0.70833059]]
A [(rad/s)/(m/s^2)]:
[[-0.00137749 -0.00837044 -0.0104128 ]
[-0.00097865  0.00045345 -0.00289436]
[ 0.00396378 -0.01579155  0.02862338]]
C_gyro_i:
[[ 0.99838819  0.0516701  -0.02347824]
[-0.05247485  0.99800657 -0.03506094]
[ 0.02161983  0.03623645  0.99910935]]
Accelerometer:
M:
[[ 0.99269657  0.          0.        ]
[-0.0037098   0.84934414  0.        ]
[ 0.00057408 -0.0280116  -0.93794705]]
```


imu0: estimated poses
0.0
0.2
0.4
0.6
0.8
1.0
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1

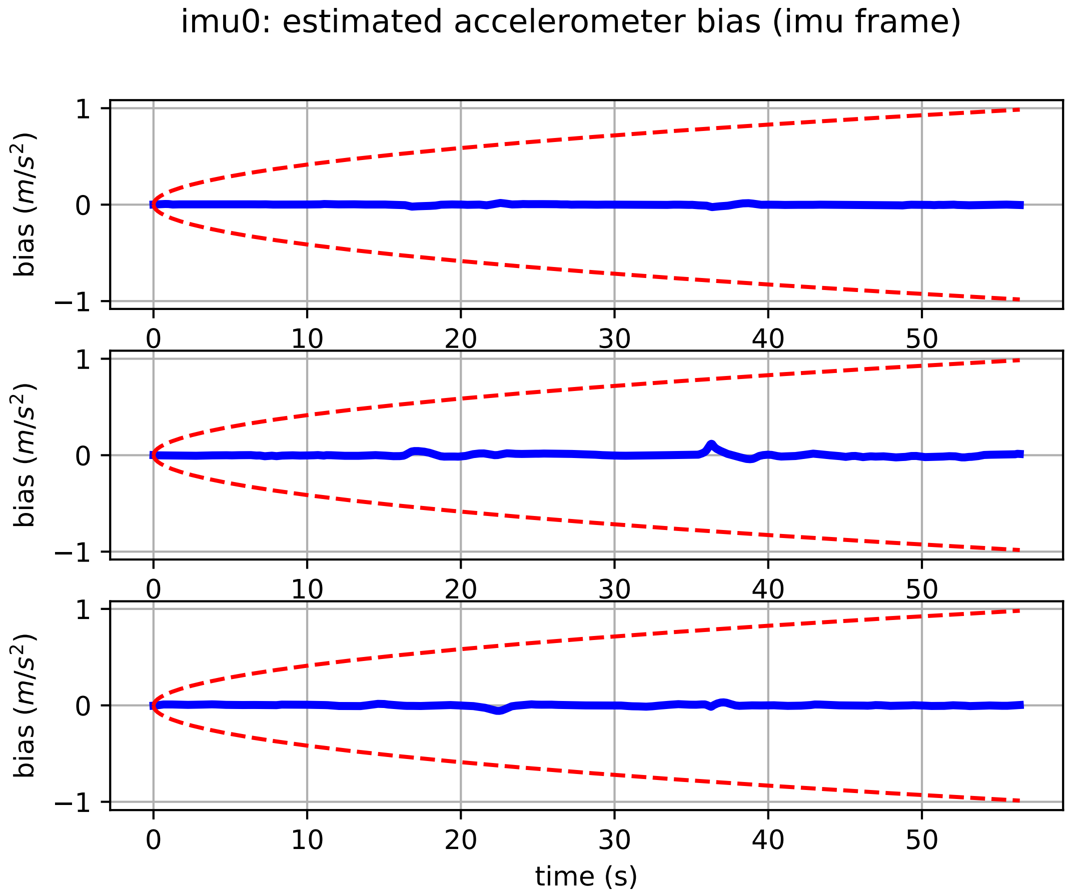

imu0: sample inertial rate
avg dt (ms) = 9.90 +- 1.9764
sample rate (ms)
time (s)
0
5
10
15
20
25
30
10
20
30
40
50



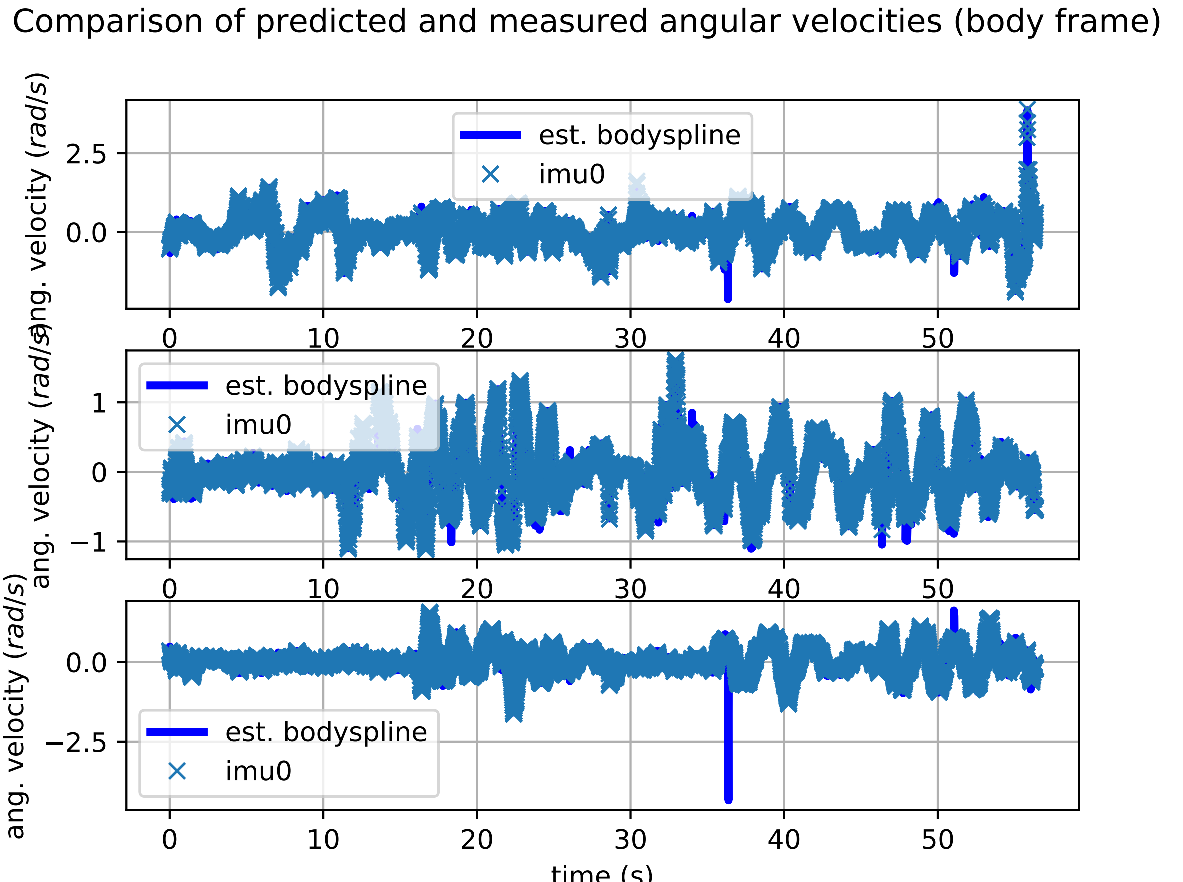
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
−8
−10
est. bodyspline
imu0
0
10
20
30
40
50
specific force (m/s^2)
0
−5
est. bodyspline
imu0
0
10
20
30
40
50
specific force (m/s^2)
0
−5
est. bodyspline
imu0
0
10
20
30
40
50
time (s)



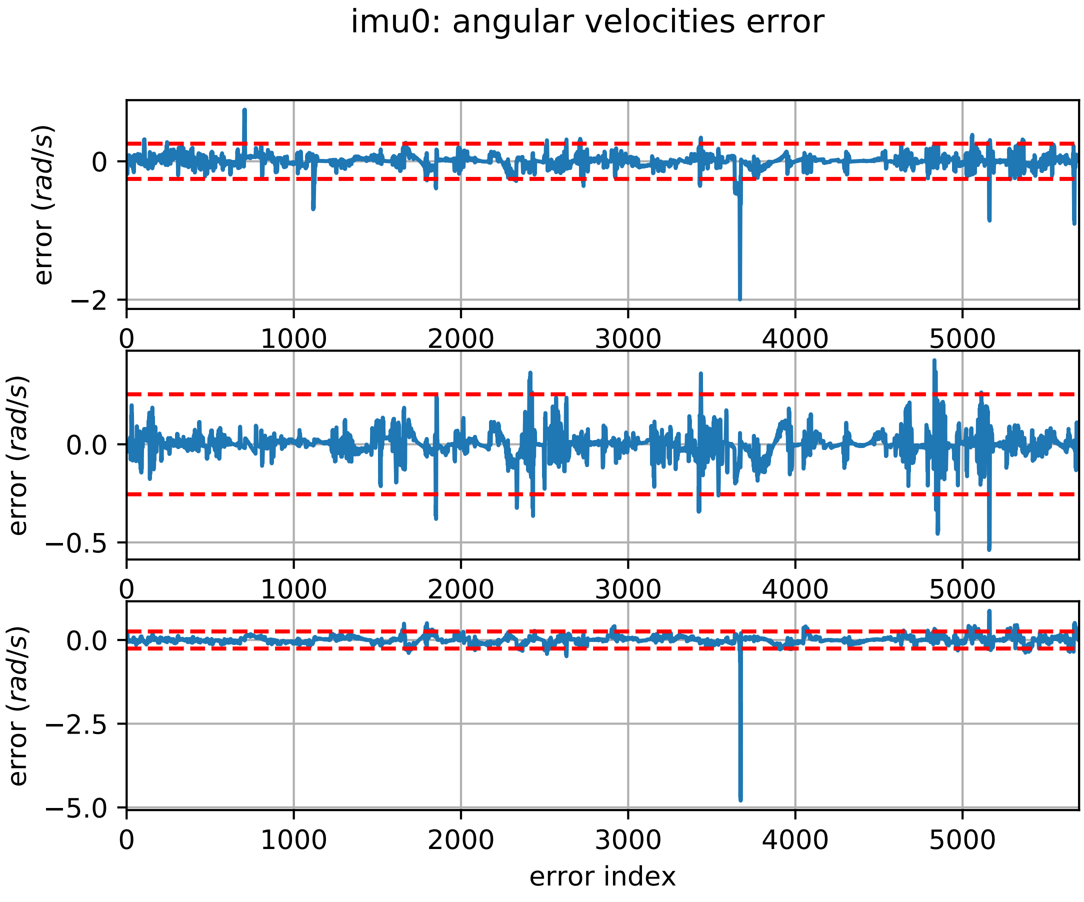
imu0: acceleration error
error (m/s^2)
1
0
−1
0
1000
2000
3000
4000
5000
error (m/s^2)
0
−5
0
1000
2000
3000
4000
5000
error (m/s^2)
0
−5
0
1000
2000
3000
4000
5000
error index



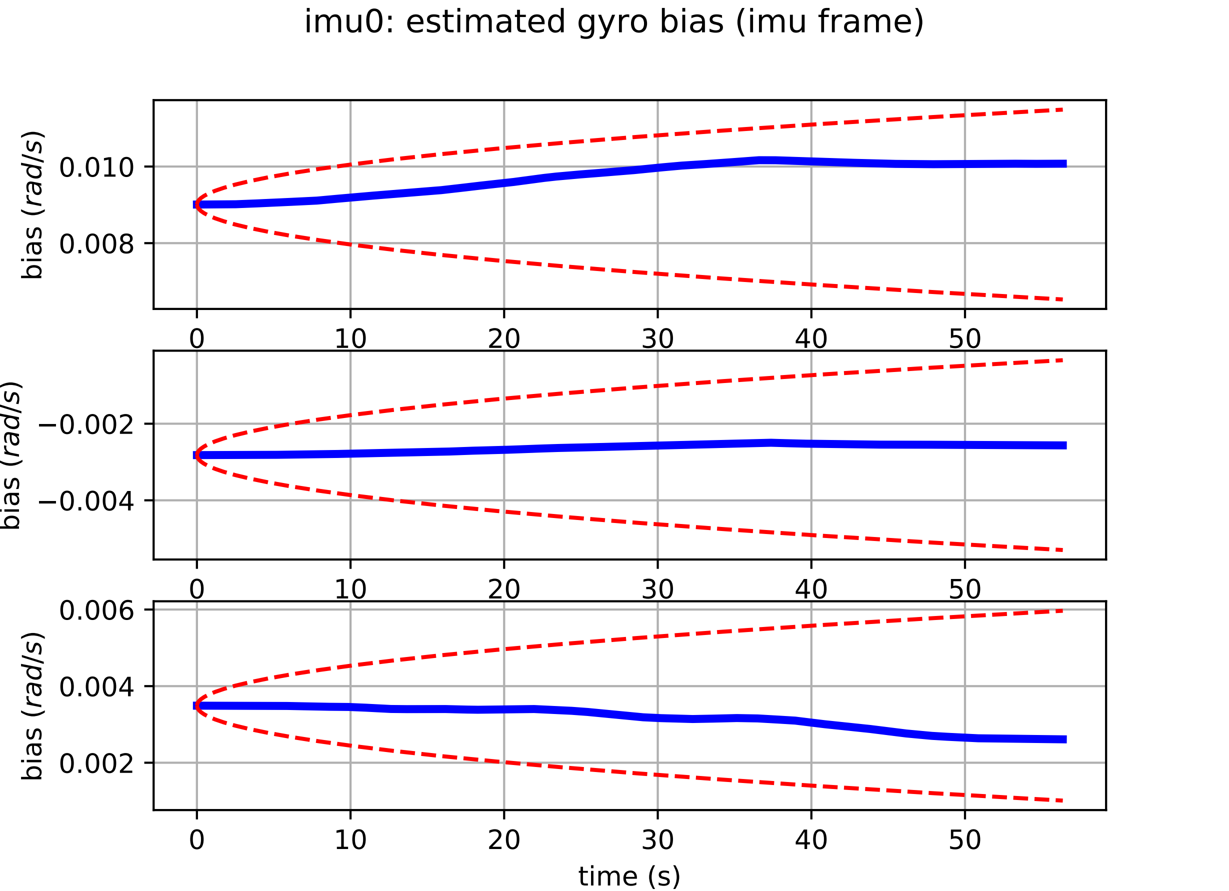
imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
1
0
−1
0
10
20
30
40
50
time (s)

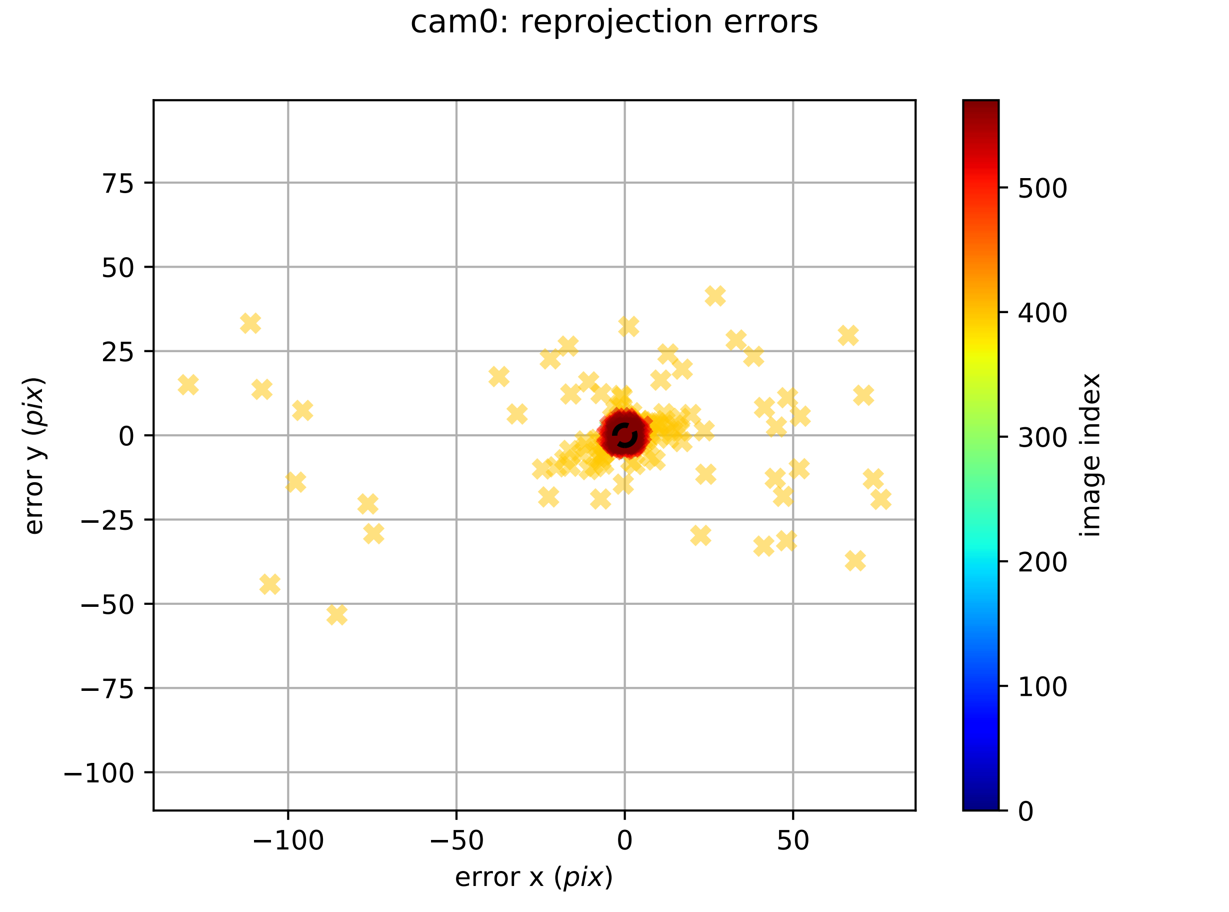


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
2.5
0.0
0
10
20
30
40
50
1
−1
−2.5
time (s)



imu0: angular velocities error
error (rad/s)
0
−2
0
1000
2000
3000
4000
5000
error (rad/s)
0.0
−0.5
error (rad/s)
0.0
−2.5
−5.0
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.008
−0.002
−0.004
0.006
0.004
0.002
0
10
20
30
40
50
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−100
−50
0
50
75
25
−25
−75
100
200
300
400
500